

Automated EDA Report ? climate_demo.csv

Source: examples/data/climate_demo.csv | Generated: 2026-06-20 15:59 UTC | Engine: polars + auto_cleaner

1. Dataset overview

Data-quality score: 100.0/100 (components: {'completeness': 100.0, 'validity': 100.0, 'uniqueness': 100.0, 'outlier_clean': 99.7})

Rows: 365 Columns: 6 In-memory: 7.2 KB Duplicate rows: 0 Warnings: 0

2. Data-health warnings

No critical data-health issues detected.

3. Auto-specialisation

Detected archetype(s): time-series (90%)

Auto-modules: inference, fda

- Time index 'date' detected ? time-series handling enabled (ordered forward-fill, functional analysis of trajectories).

4. Column profiles

Feature	Dtype	Kind	Null%	Unique	Mean	Median
date	Date	datetime	0.0%	365	-	-
temp_c	Float32	numeric	0.0%	365	14.85	15.15
humidity	Float32	numeric	0.0%	355	60.07	60.38
pressure_hpa	Float32	numeric	0.0%	365	1,013	1,013
wind_kmh	Float32	numeric	0.0%	365	12.31	12.36
is_outlier	Boolean	boolean	0.0%	2	-	-

5. Advanced analysis highlights

Significant correlations: temp_c~humidity r=-0.88

Distribution fit (example): 'temp_c' best ~ weibull_min (AIC 2,463).

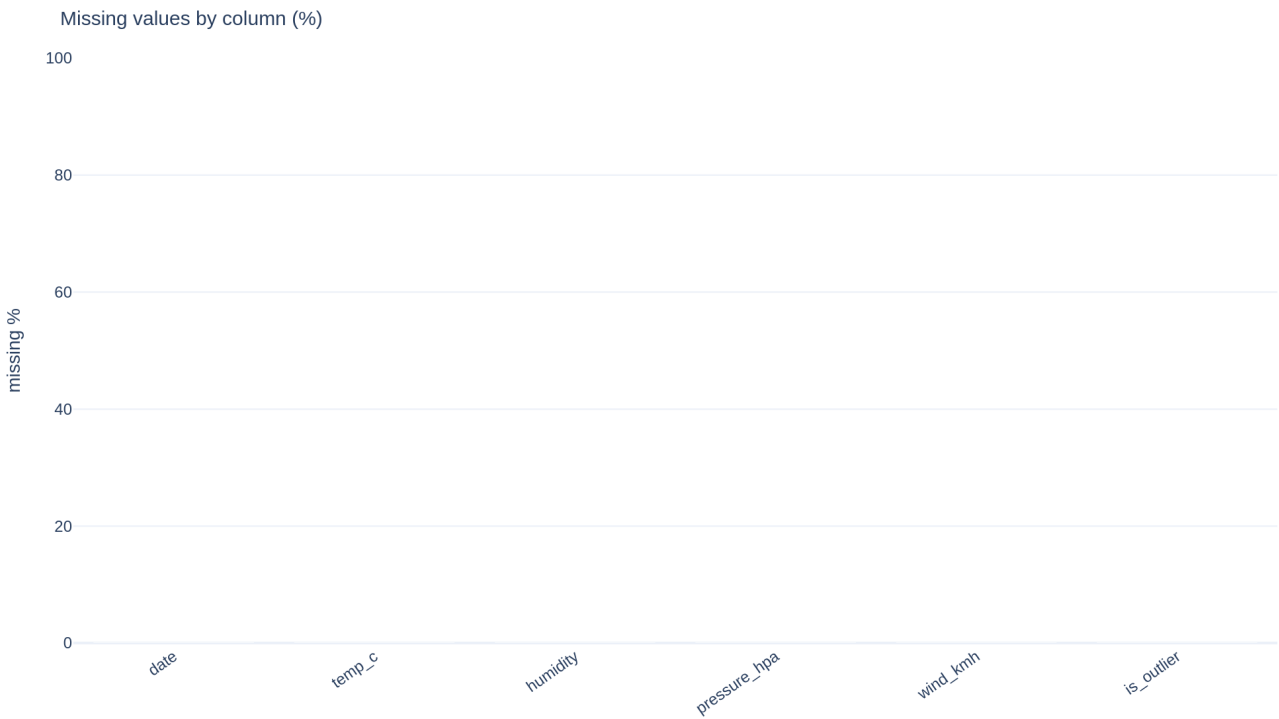
Clustering: 2 clusters (silhouette 0.333); Mahalanobis outliers: 5.

Time-series (example 'temp_c'): stationary=False, trend=decreasing, seasonal strength=0.163.

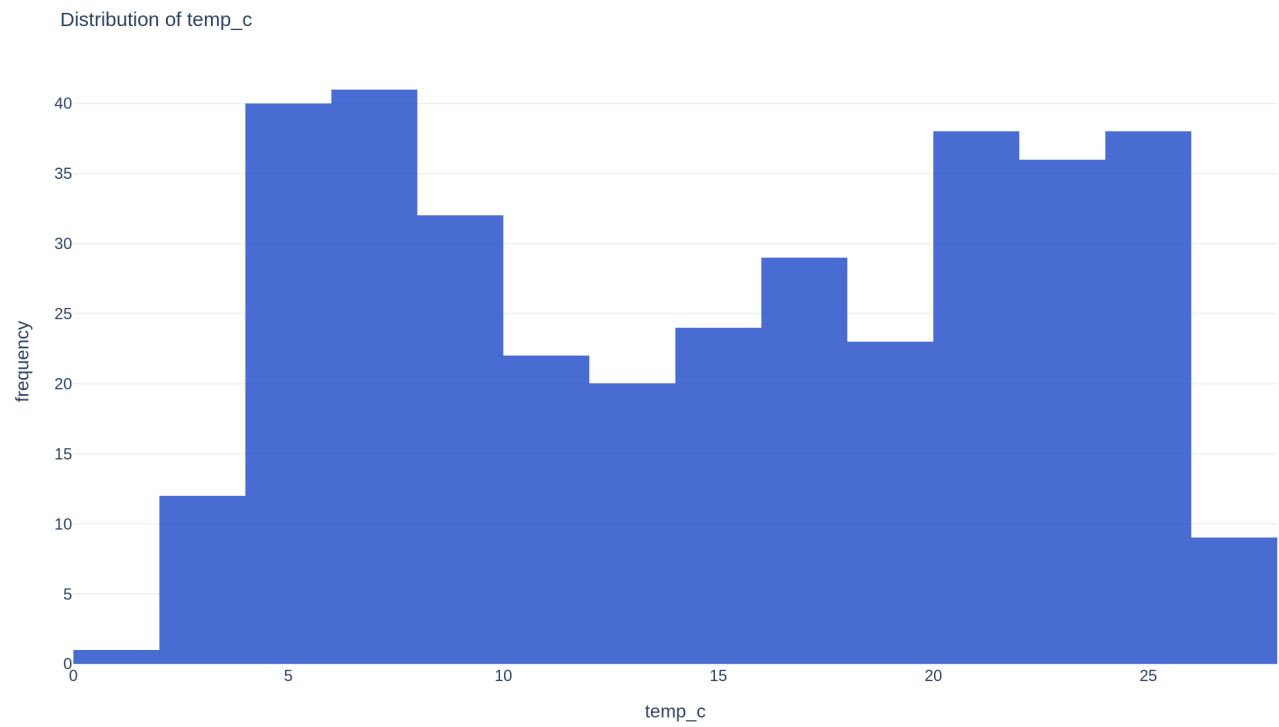
Functional PCA: 2 mode(s) capture 90% of curve variance.

6. Visualisations

Missing values by column

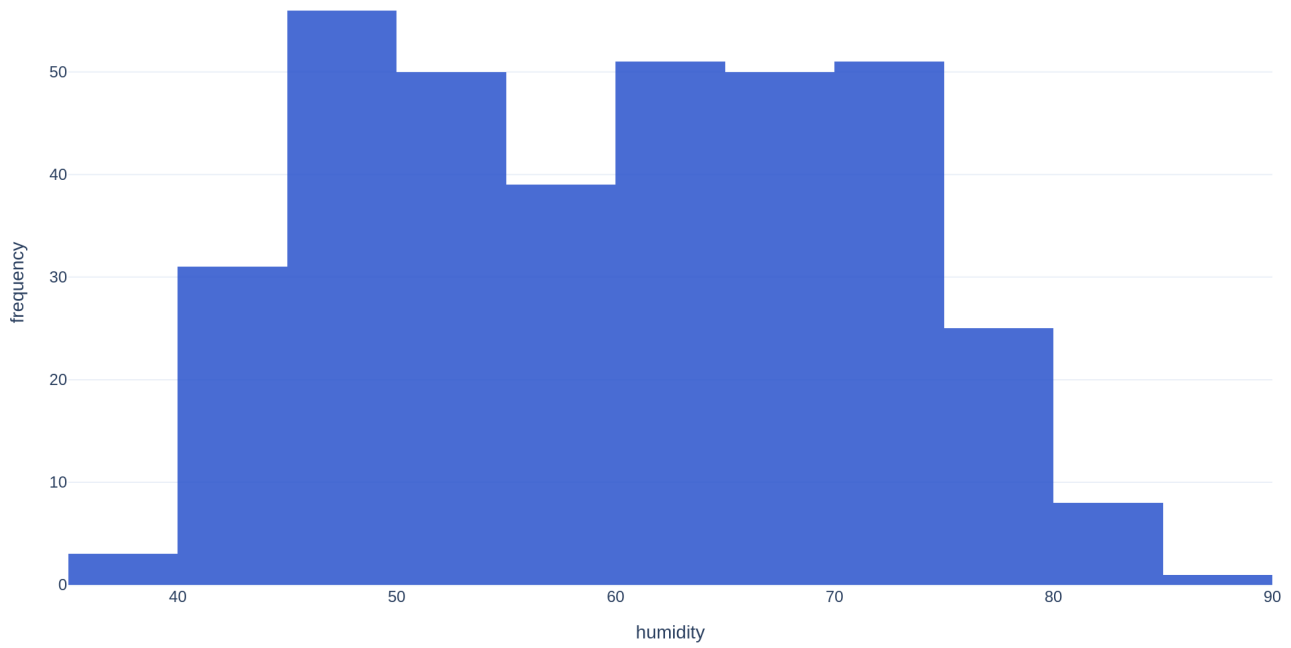


Distribution ? temp_c



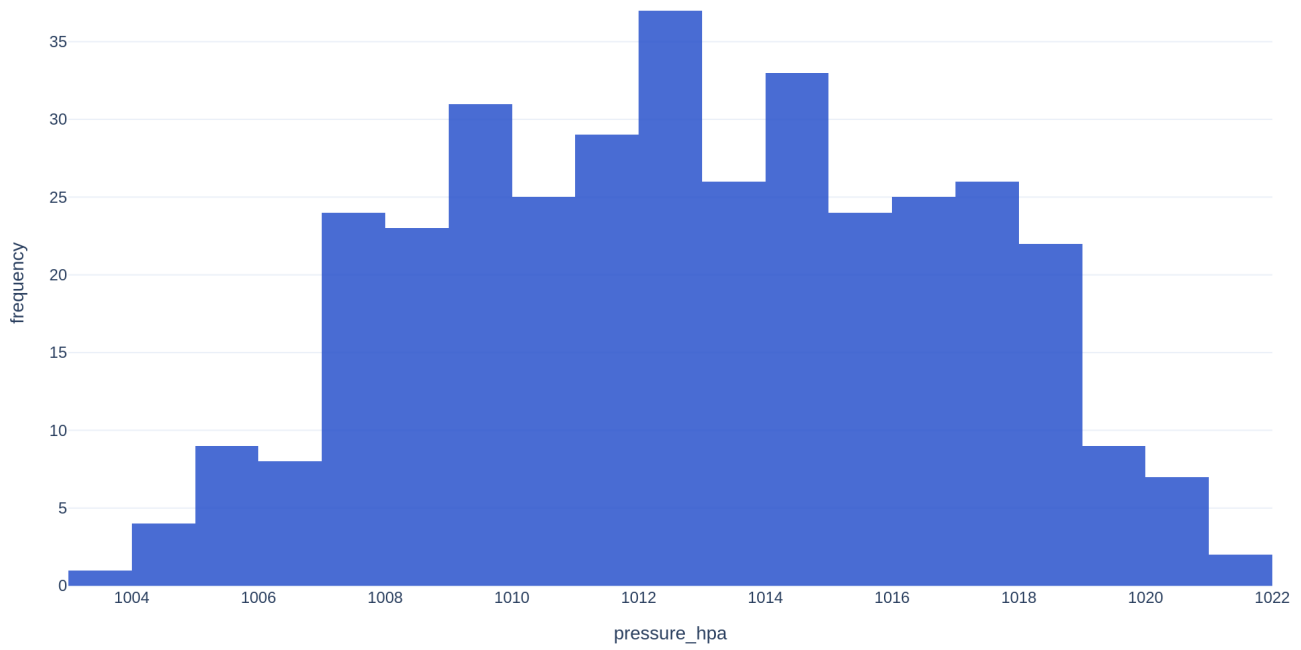
Distribution ? humidity

Distribution of humidity



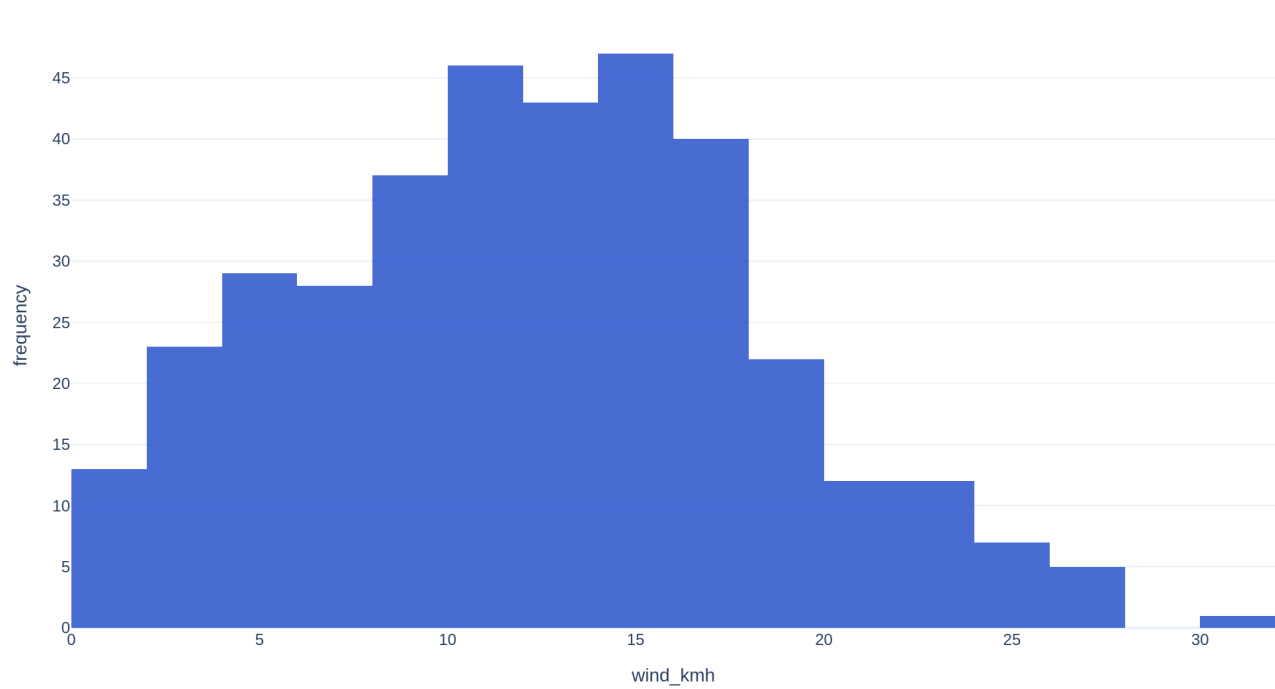
Distribution ? pressure_hpa

Distribution of pressure_hpa



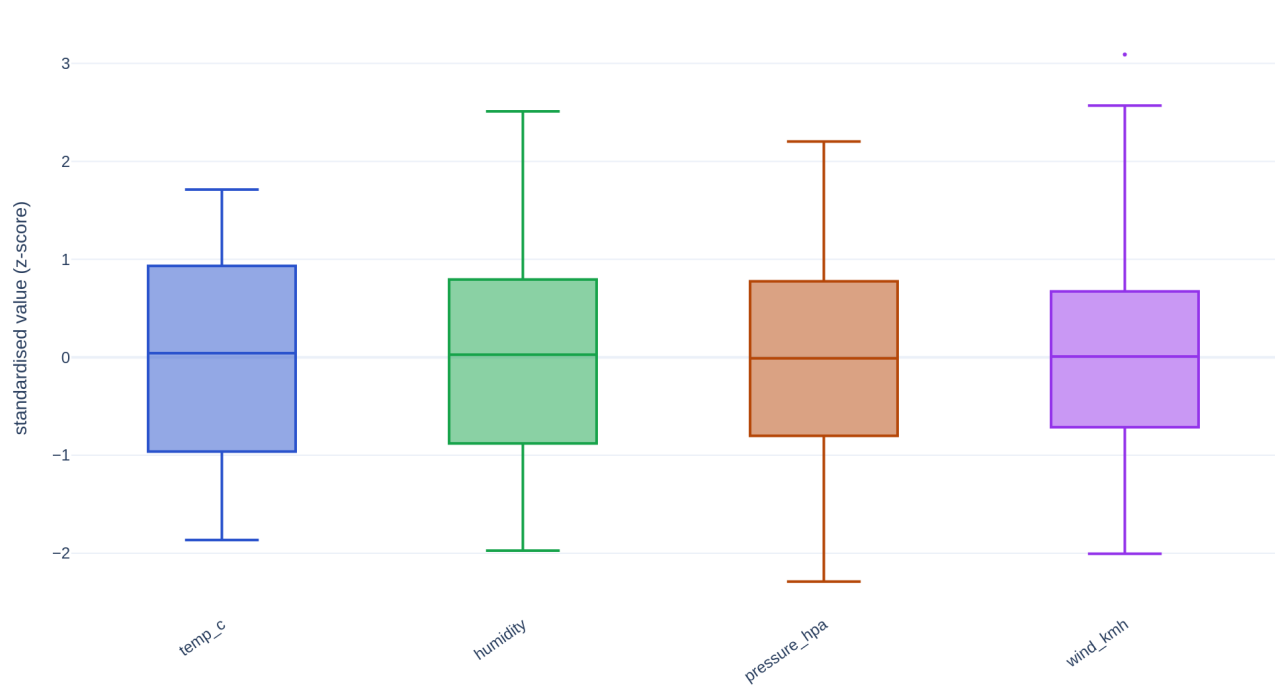
Distribution ? wind_kmh

Distribution of wind_kmh



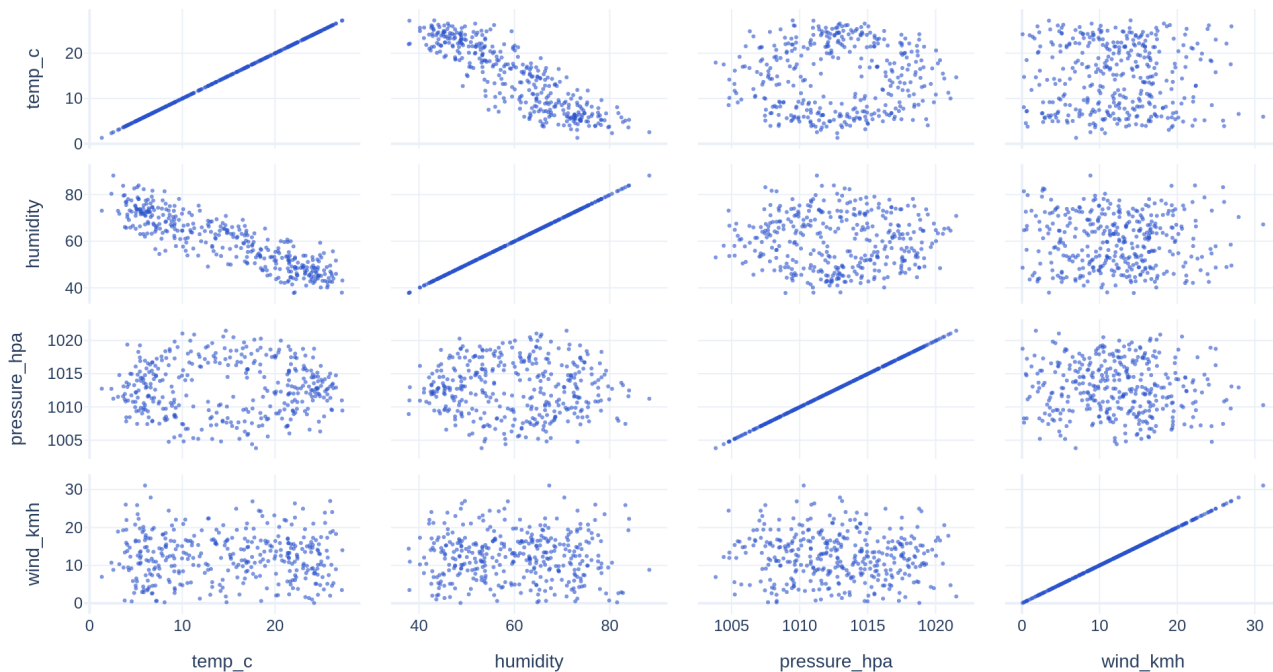
Outlier boxplots

Outlier view — standardised boxplots



Scatterplot matrix

Scatterplot matrix



Correlation heatmap

Pearson correlation heatmap

